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Roll No

EC-604 (GS)

B.E. VI Semester Examination, June 2020

Grading System (GS)

Antenna and Wave Propagation

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. Design a three element binomial array of isotropic elements positioned along the z-axis a distance 'd' apart, find the :
 - i) Normalized excitation coefficient
 - ii) Array factor
2. What do you understand by the directive gain of an antenna? Describe with a suitable expression.
3. What is Yagi Uda antenna. Design a Yagi-Uda antenna of six elements to provide a gain of 12 dBi if the operating frequency is 200 MHz.
4.
 - a) Derive an equation of potential function for sinusoidal operations of an antenna.
 - b) Write some of the practical applications of short antenna.
5. Four isotropic sources are placed symmetrically along the z-axis a distance 'd' apart. Design a binomial array find the:
 - i) Normalized excitation coefficients
 - ii) Array factor
 - iii) Angles where the array factor null occur $d = \frac{3\lambda}{4}$.
6. Prove that the Radiation Resistance of the half wave dipole is 73 ohms.
7. Two $\frac{\lambda}{2}$ dipoles are arranged broad side with a spacing of 0.67λ . The value of resistance component of mutual impedance for this spacing is 25 ohms. Calculate the ratio of power radiated by this combination to that of an isolated $\frac{\lambda}{2}$ dipole for the same input current.
8. Write short notes (any four)
 - a) MUF
 - b) LUF
 - c) Skip Distance
 - d) Parabolic Reflector Antenna
 - e) Binomial Array
 - f) Virtual height
